Wei Fu

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Education

University of Southern California
M.S. in Electrical Engineering
Northeastern University Shenyang, China
B.Eng. in Automation | GPA: 3.7545/5.0, 87.57/100 09/2021-06/2025

Research Experience

Digital Twin-based Product Quality Prediction in Industrial Processes Undergraduate Thesis
Position: Independent Research Advisor: Prof. Minghai Jiao & Chang Liu 11/2024-06/2025

1. Used thermodynamic and kinetic empirical formulas to model the effects of cold rolling and annealing on material properties.
2. Applied models like XGBoost, CatBoost, and TabTransformer, integrating physical model-generated features (e.g., grain size, dislocation density) into data-driven predictions.
3. Optimized model parameters and feature selection using evolutionary algorithms.
4. Leveraged Unity for digital twin simulation, creating an interactive UI to display process information and predict mechanical properties.

Advanced QSVM with Pinball Loss for Robust Binary Classification
Position: Research Assistant Advisor: Assistant Prof. Zheming Gao 06/2023-08/2024

1. Proposed a new model, called **Pin-QSSVM**, which integrates the **Quadratic Surface Support Vector Machine (QSSVM)** framework with the **Pinball Loss Function**. This model is designed to enhance binary classification performance by providing more flexible error penalization, particularly for datasets with imbalanced or high-noise characteristics.
2. Used **stochastic gradient descent** and **second-order cone programming** to optimize the model, testing on datasets such as **Iris, Heart Disease, German Credit Data**, and proprietary healthcare/finance data.
3. The new model achieved a **11% increase in accuracy** and **9% reduction in misclassification error** over traditional QSSVM, with a **14% improvement in F1 score** on imbalanced datasets, highlighting its robustness for high-noise applications.

Exploratory Data Analysis and Machine Learning Models for Stroke Prediction
Project-based Learning Advisor: Prof. Mark Vogelsberger 01/2023-09/2023

1. Apply Logistic Regression, Random Forest and XGBoost machine learning models to analyze and make predictions about strokes.
2. Among the tested models, Random Forest demonstrated the best performance with an accuracy of 94.14%, an F1 score of 0.9697, a recall of 0.9880, and a precision of 0.9520.
3. Logistic Regression and Decision Tree showed reasonable predictive capabilities, achieving accuracies of 72.86% and 72.14%, respectively.
4. In contrast, XGBoost performed poorly in this study, particularly with a precision of only 0.1813, although its accuracy still reached 84.85%.

Publication: Wei Fu, Exploratory Data Analysis and Machine Learning Models for Stroke Prediction, *International Conference on Data Analysis and Machine Learning*, **ISBN**: 978-989-758-705-4, DOI: 10.5220/0000185700003885 Conference Link: DAML 2023.

Development of School Bullying Intelligent Monitoring System

Position: Team Leader Advisor: Prof. Dongyue Chen 01/2022-09/2023

1. Used **YOLO v5** and **DeepSORT** to create a real-time multi-target tracking model capable of identifying and tracking individuals in crowded environments, allowing for the detection of potentially harmful interactions among students.
2. Trained the model using a combination of a public violence dataset and additional data that our team collected and organized, enhancing detection accuracy and ensuring the model effectively identifies harmful interactions relevant to middle and high school contexts.
3. Adapted the system to a wide range of cameras in primary and secondary schools to achieve intelligent monitoring and help reduce school bullying, enabling real-time alerts for school staff to facilitate prompt intervention in bullying incidents.

Internship Experience

Chongqing Jifeng Geological Exploration Co., Ltd Chongqing, China

Position: Data Engineer Intern 06/2022-09/2022

Machine Learning Applications in Lithology Classification

1. Cleaned and preprocessed Geological Mapping data to ensure accuracy and consistency, handling missing values and outliers.
2. Conducted supervised learning methods like Random Forests and Support Vector Machines (SVM) to classify different rock types and lithologies from geological data, including remote sensing, seismic, and well log data.
3. Helped in identifying lithological units and zoning in on regions with potential resource deposits.

Course Projects

Edge Computing Control System for Motor Speed Regulation 05/2024-07/2024

Course Project of Cloud Computing and Industrial Internet

Workload:

1. System Architecture Design: Developed front-end with serial and MQTT communication in PyCharm; built back-end with EdgeX Foundry, Docker, and VMware; managed container services with Docker Compose.
2. Control Algorithm Development: Implemented a PID control algorithm and a fuzzy self-adaptive PI controller; created custom Go plugins for EdgeX Foundry.
3. Real-Time Communication: Established MQTT for computer-edge server communication; configured topics and QoS; developed a monitoring system for real-time motor speed.
4. Hardware Integration: Assembled and tested hardware including a microcontroller, rectifier, and DC motor; connected and calibrated a speed sensor.

Skills & Others

Programming Skills: MATLAB/Simulink, C++, Python

Awards: Outstanding Student Award at Northeastern University, 2023

First Prize of Liaoning Mathematics Competition, 2022